

BANARAS HINDU UNIVERSITY

B.Com. (Hons.) — Semester IV (Special) Examination, 2014-15

Paper No. BCH-224: Business Mathematics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer all questions. All questions carry equal marks (14 marks each).***Question 1.**

(a) The 4th term of an A.P. is 13 and the 9th term is 28. Find the 12th term. [7]

(b) The sum of 4 terms of an A.P. is 16 and the sum of the squares of the extreme terms is 50. Find the numbers. [7]

OR(a) Find the sum of n odd natural numbers. [7](b) If a, b, c are in A.P., prove that $a^2(b+c), b^2(c+a), c^2(a+b)$ are also in A.P. [7]**Question 2.**

(a) Which term of the sequence 3, 9, 27, ... is 729? [7]

(b) Find the Geometric Progression (G.P.) whose first term is 8 and the sum of infinite terms is 16. [7]

OR(a) If a, b, c are in G.P. and x and y are the Arithmetic Means between a and b , and b and c respectively, prove that:

$$\frac{1}{x} + \frac{1}{y} = \frac{2}{b}$$

[7]

(b) If a, b, c are in A.P. and x, y, z are in G.P., prove that:

$$x^{b-c} \cdot y^{c-a} \cdot z^{a-b} = 1$$

[7]

Question 3.

- (a) If ${}^{2n+1}P_{n-1} : {}^{2n-1}P_n = 3 : 5$, find n . [7]
- (b) How many numbers can be formed between 10 and 1000 with the help of digits 2, 3, 4, 5, 0, 6, 7, 8, 9? Repetition of digits is not allowed. [7]

OR

- (a) Prove that:

$${}^{n-1}C_r + {}^{n-1}C_{r-1} = {}^nC_r$$

[7]

- (b) In how many ways can 4 men and 3 ladies be selected from 7 men and 8 ladies? [7]

Question 4.

- (a) Find the term independent of x in the expansion of $\left(x^2 + \frac{1}{x}\right)^{12}$. [7]
- (b) In the expansion of $(1+x)^{43}$, the coefficients of $(2r+1)$ th term and $(r+2)$ th term are equal. Find r . [7]

OR

- (a) Find the coefficient of x^{15} in the expansion of $(x-x^2)^{10}$. [7]
- (b) Find the 19th term of $\left(2x^{1/2} - y^{1/3}\right)^{20}$. [7]

Question 5.

- (a) If $f(x) = \frac{x-1}{x+1}$, prove that $f\left(\frac{x-1}{x+1}\right) = -\frac{1}{x}$. [4]
- (b) Evaluate:

$$\lim_{x \rightarrow 0} \frac{e^x - e^{-x}}{x}$$

[4]

- (c) Differentiate y w.r.t. x :

$$y = \frac{e^{x^3} + e^{-x^3}}{e^{x^3} - e^{-x^3}}$$

[6]

OR

- (a) Simplify:

$$\int \left(x + \frac{1}{x}\right)^2 dx$$

[4]

- (b) Evaluate the following integrals:

(i) $\int x^3 dx$

(ii) $\int x^n \log x dx$

[10]